EXAM TO WORK

2026 EDITION

# AZ-104

## The Azure Administrator Field Guide

Pass the exam, then do the job. Every skill on the official outline, explained with diagrams and real commands.

12 WORKPLACE LABS | 156 PRACTICE QUESTIONS | CLI + POWERSHELL

EVERY COMMAND, BROKEN DOWN




IDENTITY

STORAGE

COMPUTE

NETWORKING

MONITOR

JEY SINGH

2026 edition
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EXAM TO WORK

# AZ-104

## The Azure Administrator Field Guide

Pass the exam, then do the job. Every skill on the official outline, every key command broken down in Azure CLI and PowerShell, and the labs that prove you can do the work.

Identity & Governance | Storage | Compute | Networking | Monitor & Backup

**5**
exam domains, every outline bullet mapped

**2**
shells for every key task: Azure CLI and PowerShell

**12**
workplace labs plus a capstone, with scripts

**156**
original practice questions, all explained


## Jey Singh



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**No real exam questions.** Every practice question in this book is original and written from Microsoft's public skills outline. Nothing here comes from the live exam. Sharing real exam content breaks the Microsoft exam agreement and can get your certification revoked, so please never use "dumps".

**Accuracy.** Written against the official AZ-104 skills outline (skills measured as of October 26, 2023, still current in November 2026). Azure changes often. Check the study guide at aka.ms/AZ104-StudyGuide before you book. Every lab and the capstone were run end to end in a brand-new Azure subscription in November 2026 (Azure CLI 2.91 in Cloud Shell, Bicep 0.49), and fixed where Azure had changed.

**Costs.** Labs create real Azure resources, and some of them cost money. Set a budget alert first (Chapter 2) and always run the cleanup step. You are responsible for charges in your subscription.

**How this book was made.** Researched and drafted with the help of AI tools, then reviewed, tested and edited by the author.

BEFORE YOU START

# Why this book exists

Many AZ-104 resources fall into one of two traps. Some read like product documentation: accurate, but you finish a chapter without knowing what to type or what the exam will ask. Others are piles of practice questions: you memorise answers and freeze when a question is worded differently, or when your first ticket at work looks nothing like a multiple-choice question.

This book takes a third path. Every topic follows the same loop: **understand it** with a diagram and plain words, **run it** with a command broken into its parts, **test it** with exam-style questions, and **use it** in a lab written as a real work ticket.

It covers every skill on Microsoft's official outline and leaves out what the exam and the job do not need. Azure CLI and PowerShell both appear for every key task, because the exam uses both and employers ask for both.

You do not need to be an expert to start. If you can use a terminal and you know what an IP address is, you are ready. Work through one chapter at a time. In Chapters 3 to 7, read the examples to learn each command, then run the marked exercises, the mini-labs and the Chapter 8 labs. Azure CLI and PowerShell versions are alternatives: pick one shell unless an exercise asks for both. By the end you will have built, checked and cleaned up real Azure work, and you will be able to explain it in an interview.

**Good luck. You can do this.**

### Who this book is for

- Anyone preparing for the AZ-104 exam
- Help desk and support staff moving into cloud
- Students and career switchers
- Admins who want Azure skills they can prove

### What you need

- A computer with a browser
- An Azure subscription (free account works for most labs)
- About 6 weeks at 1 hour a day

### The four-step loop

**1 · Understand**
diagram + plain words

**2 · Run**
command anatomy

**3 · Test**
quiz + practice exam

**4 · Use**
ticket labs

INSIDE

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Five exam domains, the commands that matter, and the practice that gets you hired.


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## How to use this book

1. **Read one topic**
   Each page maps to a line of the official outline. Look for the purple tags.
2. **Learn the command, then run the labs**
   Every key command is broken into parts and shown in Azure CLI and PowerShell (pick one). Run things when a mini-lab or Chapter 8 lab tells you to.
3. **Do the quiz, then the ticket lab**
   Chapters 3 to 7 end with 10 questions. Chapter 8 turns the skills into real work tickets.
4. **Sit the practice exam**
   Use 80% as a revision target, then test yourself on fresh questions before you book.

**ON THE EXAM**
What Microsoft tests and how questions are worded.

**EXAM TRAP**
The wrong answer that looks right.

**AT WORK**
How this shows up in a real ticket.

**MEMORY HOOK**
A trick to remember it for good.

**TRY IT NOW**
A 5 to 10 minute hands-on exercise.

**COST WARNING**
This one can charge your card. Clean up.

START HERE

# Choose your route

AZ-104 is an intermediate exam. Microsoft expects you to know the basics of operating systems, networking and virtualization. Take the five-minute check, pick a route, and switch routes any time.

**FOUNDATION CHECK (5 MINUTES, NO LOOKING THINGS UP)**

1. What is the difference between a tenant, a subscription and a resource group?
2. What does `10.0.1.0/24` mean, and roughly how many addresses does it hold?
3. What is the difference between a public IP address and a private one?
4. What does DNS do?
5. In a terminal, how do you create a text file, edit it and save it?

Answers: 1. Tenant = your organisation's Entra ID directory; subscription = a billing and access boundary inside it; resource group = a folder for resources that live and die together. 2. A network of 256 addresses (Azure keeps 5 in every subnet). 3. Public is reachable from the internet; private only inside a network. 4. Turns names into IP addresses. 5. In Cloud Shell, `code notes.txt` opens an editor; Ctrl+S saves. In a command, `@notes.json` means "read this file". **Four or more right:** route 2 or 3. **Fewer:** start with route 1.

| Route | Order | Move on when |
|---|---|---|
| **1. New to cloud**<br>Little IT background | Chapter 1, then Chapter 2: set up the safe lab account and run the safe lab loop twice. Fill gaps with Microsoft Learn's free Azure Fundamentals (AZ-900) modules. Then Chapters 3 to 7, one at a time, each with its mini-lab. Start Chapter 8 with labs 01, 03 and 09, then the rest. | You pass the foundation check, can run the safe lab loop from memory, and score 7/10 or more on each chapter quiz. |
| **2. First AZ-104 attempt**<br>Some IT experience | Skim Chapters 1 and 2. After each of Chapters 3 to 7, do its Chapter 8 labs: Identity → 01, 02 · Storage → 03, 04 · Compute → 05 to 08 · Networking → 09, 10 · Monitor → 11, 12. Then Practice exam 1, the capstone, and Practice exam 2 a week later. | You can finish a lab's "try it yourself" without the solution, and score 80% on Practice exam 1 with every miss explained. |
| **3. Working admin**<br>Uses Azure already | Sit Practice exam 1 first as a diagnostic. Read only your weak domains, run the labs for tasks you have not done for real, then build the capstone and grade it with the handover checklist. Finish with Practice exam 2. | Every weak domain is above 80%, and your capstone passes the handover checklist. |

**FOR EVERY ROUTE**

Two practice exams, and they do different jobs. Practice exam 1 is for learning: retake it, explain every miss. Practice exam 2 is for checking: sit it once, cold, when you think you are ready. Your score on fresh questions tells you more than a repeat of the same 56. The separate Practice Pack adds 151 more questions, including a third exam.

**STUCK ON A WORD OR AN ERROR?**

Chapter 12 is the reference: where each task sits in the portal, 28 real error messages with fixes, a glossary and an index. Use it whenever a chapter or a lab stops making sense.

FOUNDATIONS · CHAPTER

# 02

# Your Toolkit: Command Grammar

**Used in every domain**

Azure CLI and Azure PowerShell look like hundreds of commands. They are really two short sentence patterns. Learn the patterns once and you can read, guess and write almost any command, on the exam and at work.

---

YOU WILL BE ABLE TO

- Set up budget alerts and a cleanup habit that keep lab costs small
- Sign in with Cloud Shell, the CLI and PowerShell
- Break any az command into its parts
- Break any Az PowerShell cmdlet into its parts
- Translate 25 everyday tasks between CLI and PowerShell
- Filter output with --query and use variables in scripts

BEFORE YOUR FIRST COMMAND

# Set up a safe lab account

Ten minutes now saves you from the classic beginner story: "I left a VM running and got a bill."



### 1 Get a subscription

Students: **Azure for Students** gives a yearly credit with no credit card. Everyone else: the **Azure free account** gives a credit for the first 30 days plus some always-free services. At work you will usually get a sandbox subscription from your team.

### 2 Set a budget alert first

Portal: search **Budgets** (under Cost Management), choose your subscription, **Add**. Set a monthly amount of **US$20** (or the same in your currency) and add email alerts at 50%, 80% and 100%. A budget only **warns**; it does not stop resources. Deleting the lab is what stops cost.

### 3 Open Cloud Shell

Click the `>_` icon at the top of the portal. Choose **Bash** for Azure CLI or **PowerShell** for Az PowerShell. Both tools are already installed and you are already signed in. If asked, pick "No storage account required" for a quick session, or mount storage to keep your files.

### 4 Pick one region, then check it

This book uses `canadacentral`, with `canadaeast` as the second region. Use one close to you and keep everything there. New subscriptions are often blocked from small VM sizes in busy regions, so before the first VM lab run `bash tools/check-region.sh` from the lab repo (Chapter 8). When we tested a brand-new account, it moved us from Canada Central to Canada East.

AZURE CLI · OPTIONAL (SKIP IT IF YOU RAN TOOLS/SETUP-SUBSCRIPTION.SH)

```
az consumption budget create --budget-name lab-budget-cli --amount 20 \
    --category cost --time-grain monthly \
    --start-date $(date +%Y-%m-01) \
    --end-date $(date -d "+1 year" +%Y-%m-01)
# $( ) runs a command and pastes its result, here the 1st of this month
# Add the email alerts in the portal: Budgets > lab-budget-cli > Alert conditions
```

COST WARNING: THE THREE THINGS THAT COST MONEY WHILE IDLE

- **Running VMs.** "Stopped" inside the OS still bills compute. Use **Stop** in the portal or `az vm deallocate`.
- **Gateways and firewalls.** VPN gateways, Bastion, Azure Firewall and Application Gateway bill by the hour even with zero traffic. Create, test, delete the same day.
- **Disks and public IPs** left behind after you delete a VM. That is why every lab in this book lives in its own resource group that you delete at the end.

## Working on your own computer

AZURE CLI · INSTALL AND SIGN IN

```
# Windows
winget install -e --id Microsoft.AzureCLI
# macOS
brew install azure-cli
# then
az login
az account show -o table
EnvironmentName    IsDefault    Name    State    TenantId               (trimmed)
-----------------  -----------  ------  -------  -------------------------------------
AzureCloud         True         Lab     Enabled  00000000-0000-0000-0000-000000000000
```

AZURE POWERSHELL · INSTALL AND SIGN IN

```
# PowerShell 7, then the Az module
Install-Module -Name Az -Repository PSGallery -Scope CurrentUser
Connect-AzAccount
Get-AzContext
```

`az login` opens a browser. On a machine without a browser use `az login --use-device-code`, then enter the code on another device. If you have more than one subscription, always check which one is active before you create anything:

AZURE CLI

```
az account list -o table
az account set --subscription "Lab"
```

AZURE POWERSHELL

```
Get-AzSubscription
Set-AzContext -Subscription "Lab"
```

**AT WORK**

The most expensive mistake juniors make is running a command against the **wrong subscription**, for example production instead of dev. Make `az account show` a reflex before any create or delete.

PATTERN 1

# Azure CLI grammar

Every Azure CLI command is the same sentence: **az**, then **what** you are working on, then the **action**, then the **details**.

COMMAND ANATOMY · AZURE CLI




**Say it:** "Azure, in networking, virtual networks: create one in the resource group rg-lab02, call it vnet-hub, and give it the IP range 10.10.0.0/16."

**Colour key:** grey = the tool · coloured = what you work on · orange = the action · blue = an option (flag) that asks a question · light box = your answer to it, which you type.

WHAT YOU'LL SEE (TRIMMED)

```
{
  "newVNet": {
    "addressSpace": { "addressPrefixes": [ "10.10.0.0/16" ] },
    "location": "canadacentral",
    "name": "vnet-hub",
    "provisioningState": "Succeeded",
    "resourceGroup": "rg-lab02",
    ...
  }
}
```

Look for **"provisioningState": "Succeeded"**. That means Azure built it. Every create command in this book returns JSON like this.

### Groups nest, like folders

The group can be one word or several. Each extra word narrows it down:

FROM BROAD TO SPECIFIC

```
az network
az network vnet
az network vnet subnet
az network vnet subnet create
```

Type any partial command with `--help` to see what lives under it: `az network vnet --help`.

### The five verbs you use every day

| Verb | Meaning |
|---|---|
| `list` | Show me all of them |
| `show` | Show me this one, in detail |
| `create` | Make a new one |
| `update` | Change settings on an existing one |
| `delete` | Remove it |

Special verbs exist where they make sense: `start`, `stop`, `deallocate`, `resize`, `upload`, `swap`, `renew`.

## Short flags save typing

| Long form | Short form | Meaning |
|---|---|---|
| `--resource-group` | `-g` | Which resource group |
| `--name` | `-n` | Name of the resource |
| `--location` | `-l` | Azure region |
| `--output` | `-o` | `json` (default), `table`, `tsv`, `yaml`, `none` |
| `--subscription` | (none) | Run against a different subscription for this one command |

**MEMORY HOOK: G-N-L**

Almost every create command needs **G**roup, **N**ame, **L**ocation: `-g -n -l`. Think "Go Name Location". Then add the one or two settings that make this resource different (a SKU, an address range, an image).

**ON THE EXAM**

Hot-area questions show a command with blanks, for example `az storage account create ... --sku ____`. You do not need to memorise every parameter. You need to recognise **which group** owns the task and **which value** meets the requirement (here: the redundancy option). Train this by reading every anatomy block in this book out loud.

**AT WORK: EXTENSIONS AND PREVIEW COMMANDS**

Some CLI groups are extensions that install the first time you use them, for example `az network bastion`, `az dataprotection` and `az monitor log-analytics query`. If the CLI asks to install an extension, say yes. In scripts, run `az config set extension.use_dynamic_install=yes_without_prompt` once.

PATTERN 2

# Azure PowerShell grammar

PowerShell flips the order: the **action** comes first, joined by a dash to **Az** plus the **noun**. Then come the details as named parameters.

COMMAND ANATOMY · AZURE POWERSHELL (AZ MODULE)




**Say it:** "New Azure virtual network in rg-lab02, named vnet-hub, in Canada Central, with 10.10.0.0/16."

### The verbs map cleanly to the CLI

| PowerShell verb | CLI verb |
|---|---|
| `Get-` | `list` / `show` |
| `New-` | `create` |
| `Set-` / `Update-` | `update` |
| `Remove-` | `delete` |
| `Start-` / `Stop-` | `start` / `stop`, `deallocate` |
| `Add-` | attach something to something |

### Objects, not text

The CLI returns text (JSON). PowerShell returns **objects** you can store, change and pass on with the pipe `|`. That is why many PowerShell changes follow a **get, change, save** pattern:

AZURE POWERSHELL · RESIZE A VM

```
$vm = Get-AzVM -ResourceGroupName rg-lab05 -Name vm-web01
$vm.HardwareProfile.VmSize = "Standard_B2als_v2"
Update-AzVM -VM $vm -ResourceGroupName rg-lab05
```

The CLI does the same in one line: `az vm resize -g rg-lab05 -n vm-web01 --size Standard_B2als_v2`.

**MEMORY HOOK: "VERB-AZ-NOUN, SINGULAR"**

PowerShell nouns are always **singular**: `Get-AzVM`, not Get-AzVMs, even when it returns fifty VMs. If you can guess the noun, you can find the cmdlet: `Get-Command -Noun AzVirtualNetwork*`. Then `Get-Help New-AzVM -Examples` shows real usage.

**ON THE EXAM**

Test-takers report many more Azure CLI questions than PowerShell ones, but PowerShell still appears. The skill tested is the same: pick the right cmdlet and the right parameter value. If you know the CLI version, use the verb table above to translate.



ROSETTA STONE

## 25 everyday tasks, both ways

Variables used below: `$RG` resource group, `$LOC` region. Each row is a real command you will meet again later in the book.

| Task | Azure CLI | Azure PowerShell |
|---|---|---|
| **Sign in** | `az login` | `Connect-AzAccount` |
| **Choose subscription** | `az account set --subscription "Lab"` | `Set-AzContext -Subscription "Lab"` |
| **Create resource group** | `az group create -n $RG -l $LOC` | `New-AzResourceGroup -Name $RG -Location $LOC` |
| **List resource groups** | `az group list -o table` | `Get-AzResourceGroup` |
| **Delete resource group** | `az group delete -n $RG --yes --no-wait` | `Remove-AzResourceGroup -Name $RG -Force -AsJob` |
| **Create user** | `az ad user create --display-name "Ana Lee" --user-principal-name ana@contoso.com --password '...'` | `New-AzADUser -DisplayName "Ana Lee" -UserPrincipalName ana@contoso.com -Password $pw -MailNickname ana` |
| **Create group** | `az ad group create --display-name Helpdesk --mail-nickname helpdesk` | `New-AzADGroup -DisplayName Helpdesk -MailNickname helpdesk` |
| **Assign a role** | `az role assignment create --assignee ana@contoso.com --role Reader --scope /subscriptions/<id>/resourceGroups/$RG` | `New-AzRoleAssignment -SignInName ana@contoso.com -RoleDefinitionName Reader -ResourceGroupName $RG` |
| **Who has access?** | `az role assignment list -g $RG -o table` | `Get-AzRoleAssignment -ResourceGroupName $RG` |
| **Add a delete lock** | `az lock create -n no-delete --lock-type CanNotDelete -g $RG` | `New-AzResourceLock -LockName no-delete -LockLevel CanNotDelete -ResourceGroupName $RG` |
| **Add a tag** | `az tag update --resource-id $ID --operation Merge --tags env=dev` | `Update-AzTag -ResourceId $ID -Tag @{env="dev"} -Operation Merge` |
| **Storage account** | `az storage account create -n stlab01 -g $RG -l $LOC --sku Standard_LRS` | `New-AzStorageAccount -ResourceGroupName $RG -Name stlab01 -Location $LOC -SkuName Standard_LRS` |
| **Blob container** | `az storage container create -n docs --account-name stlab01 --auth-mode login` | `New-AzStorageContainer -Name docs -Context $ctx` |

| | | |
|---|---|---|
| **Upload a file** | `az storage blob upload --account-name stlab01 -c docs -n a.txt -f ./a.txt --auth-mode login` | `Set-AzStorageBlobContent -File ./a.txt -Container docs -Blob a.txt -Context $ctx` |
| **Show access keys** | `az storage account keys list -n stlab01 -g $RG` | `Get-AzStorageAccountKey -ResourceGroupName $RG -Name stlab01` |
| **Virtual network** | `az network vnet create -n vnet-hub -g $RG --address-prefixes 10.10.0.0/16 --subnet-name snet-web --subnet-prefixes 10.10.1.0/24` | `New-AzVirtualNetwork -Name vnet-hub -ResourceGroupName $RG -Location $LOC -AddressPrefix 10.10.0.0/16` |
| **Network security group** | `az network nsg create -n nsg-web -g $RG` | `New-AzNetworkSecurityGroup -Name nsg-web -ResourceGroupName $RG -Location $LOC` |
| **Peer two VNets** | `az network vnet peering create -n hub-to-spoke -g $RG --vnet-name vnet-hub --remote-vnet vnet-spoke1 --allow-vnet-access` | `Add-AzVirtualNetworkPeering -Name hub-to-spoke -VirtualNetwork $hub -RemoteVirtualNetworkId $spoke.Id` |
| **Create a VM** | `az vm create -g $RG -n vm-web01 --image Ubuntu2204 --size Standard_B2ats_v2 --admin-username azureuser --generate-ssh-keys` | `New-AzVM -ResourceGroupName $RG -Name vm-web01 -Location $LOC -Image Ubuntu2204 -Size Standard_B2ats_v2 -Credential (Get-Credential)` |
| **Stop paying for a VM** | `az vm deallocate -g $RG -n vm-web01` | `Stop-AzVM -ResourceGroupName $RG -Name vm-web01 -Force` |
| **Resize a VM** | `az vm resize -g $RG -n vm-web01 --size Standard_B2als_v2` | Get, change `HardwareProfile.VmSize`, `Update-AzVM` |
| **Deploy Bicep/ARM** | `az deployment group create -g $RG --template-file main.bicep` | `New-AzResourceGroupDeployment -ResourceGroupName $RG -TemplateFile main.bicep` |
| **Web app** | `az webapp create -g $RG -p plan-web -n app-demo01 --runtime "NODE:22-lts"` | `New-AzWebApp -ResourceGroupName $RG -Name app-demo01 -AppServicePlan plan-web -Location $LOC` |
| **Read a metric** | `az monitor metrics list --resource $VMID --metric "Percentage CPU"` | `Get-AzMetric -ResourceId $VMID -MetricName "Percentage CPU"` |
| **Recovery Services vault** | `az backup vault create -n rsv-lab -g $RG -l $LOC` | `New-AzRecoveryServicesVault -Name rsv-lab -ResourceGroupName $RG -Location $LOC` |

`--auth-mode login` uses your Entra ID sign-in, which needs a data role such as Storage Blob Data Contributor on the account (being Owner is not enough, see Chapter 4). Without it, use `--auth-mode key`.

**EXAM TRAP**

`Stop-AzVM` **deallocates** by default (you stop paying for compute). Add `-StayProvisioned` and it only stops the OS, and you keep paying. In the CLI it is the reverse: `az vm stop` keeps it allocated, `az vm deallocate` releases it.

READING RESULTS

# Output, --query and variables

Real admin work is mostly reading. These tricks turn walls of JSON into the one value you need.



## Choose the output format

AZURE CLI

```
az vm list -o table          # human-friendly columns
az vm list -o json           # everything (default)
az vm list -o tsv            # plain values for scripts, no quotes
```

WHAT -O TABLE LOOKS LIKE

```
Name       ResourceGroup    Location       Zones
---------  ---------------  -------------  -------
vm-web01   RG-LAB05         canadacentral
vm-web02   RG-LAB05         canadacentral  1
```

## Filter with --query (JMESPath)

QUERY ANATOMY

```
[?location=='canadacentral']  .  {Name:name, Size:hardwareProfile.vmSize}
```

- `[?location=='canadacentral']` — keep only items in Canada Central
- `.` — then
- `{Name:name, Size:hardwareProfile.vmSize}` — show these two fields, with new column names

**Say it:** "From the list, keep the items in Canada Central, then show only their name and size."

AZURE CLI · FIVE QUERIES WORTH KEEPING

```
az group list --query "[].name" -o tsv
az vm list --query "[?location=='canadacentral'].{Name:name, Size:hardwareProfile.vmSize}" -o table
az vm list -d --query "[?powerState=='VM running'].name" -o tsv   # -d adds power state
az storage account list --query "[].{Name:name, Sku:sku.name}" -o table
az account show --query id -o tsv                                 # just the subscription ID
```

WHAT THE SECOND QUERY PRINTS

```
Name       Size
---------  -----------------
vm-web01   Standard_B2ats_v2
vm-web02   Standard_B2als_v2
```

PowerShell filters objects with `Where-Object` and picks fields with `Select-Object`:

AZURE POWERSHELL

```
Get-AzVM -Status | Where-Object PowerState -eq "VM running" | Select-Object Name, ResourceGroupName
Get-AzStorageAccount | Select-Object StorageAccountName, @{n="Sku";e={$_.Sku.Name}}
```

## Variables and line breaks

BASH (CLOUD SHELL)

```
RG=rg-lab02
LOC=canadacentral
VMID=$(az vm show -g $RG -n vm-web01 \
  --query id -o tsv)
echo $VMID
# line continues with a backslash \
```

POWERSHELL

```
$RG  = "rg-lab02"
$LOC = "canadacentral"
$vm  = Get-AzVM -ResourceGroupName $RG `
  -Name vm-web01
$vm.Id
# line continues with a backtick `
```

**EXAM TRAP AND WORK TRAP**

Bash variables have no `$` when you set them ( `RG=rg-lab02` ) and no spaces around `=`. PowerShell variables always have `$`. Mixing them up is the number one reason copied scripts fail.

## Never guess: ask the tool

| Question | Azure CLI | Azure PowerShell |
|---|---|---|
| What commands exist? | `az network --help` | `Get-Command -Module Az.Network` |
| Show me an example | `az find "vm create"` | `Get-Help New-AzVM -Examples` |
| What parameters? | `az vm create --help` | `Get-Help New-AzVM -Parameter *` |
| Set defaults | `az configure --defaults group=$RG location=$LOC` | `$PSDefaultParameterValues` |

### HOW EVERY LAB WORKS

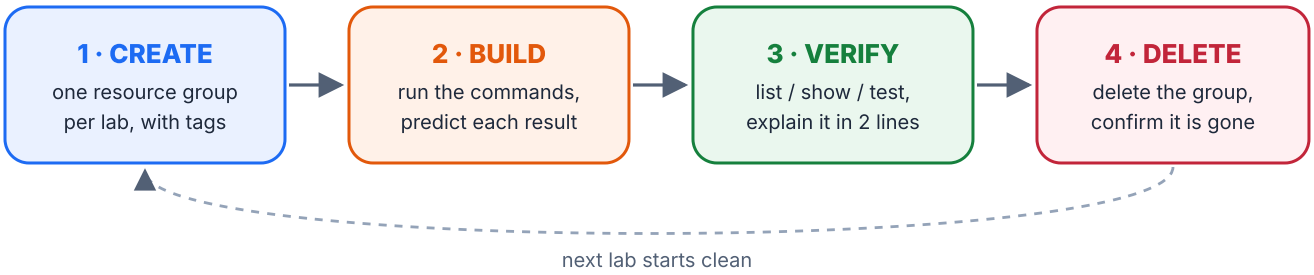
**HOW TO USE THE EXAMPLES IN CHAPTERS 3 TO 7**

Each example shows **one task**, and many build on a VNet, VM or account from earlier in that chapter. They are for learning the command, not a script to paste top to bottom. To build something from zero, use the chapter **mini-lab** or the tested labs in Chapter 8. Azure CLI and PowerShell versions are alternatives: run one, not both.

**Reading a code block:** bright lines are commands. Grey lines after a command are sample output: do not type them. A line ending in `\` (bash) or `` ` `` (PowerShell) continues on the next line, so copy both. Text after `#` is a comment.

# The safe lab loop

Every hands-on exercise in this book follows the same four steps. Make it a habit, and the last step proves nothing is left running.




**AZURE CLI · THE LOOP AS COMMANDS**

```
# 1 CREATE
RG=rg-lab02; LOC=canadacentral
az group create -n $RG -l $LOC --tags owner=alex purpose=lab
# 2 BUILD (example)
az network vnet create -g $RG -n vnet-hub --address-prefixes 10.10.0.0/16
# 3 VERIFY: stop here, look at the result, then go on
az resource list -g $RG -o table
# 4 DELETE everything in one go
az group delete -n $RG --yes --no-wait
az group list --tag purpose=lab -o table     # anything left behind?
```

**WHAT STEP 3 SHOWS (SAMPLE OUTPUT, DO NOT TYPE)**

```
Name      ResourceGroup    Location       Type                                 Status
--------  ---------------  -------------  -----------------------------------  --------
vnet-hub  rg-lab02         canadacentral  Microsoft.Network/virtualNetworks
```

**TRY IT NOW (5 MINUTES, FREE)**

1. Open Cloud Shell (Bash) and run steps 1 to 3 above, one line at a time.
2. Before step 3, predict what `az resource list` will show. Were you right? Then run step 4 and check nothing is left.
3. Repeat it in Cloud Shell (PowerShell) using the right-hand column of the Rosetta table: `New-AzResourceGroup`, `New-AzVirtualNetwork`, `Get-AzResource -ResourceGroupName`, `Remove-AzResourceGroup`.

A resource group and an empty VNet cost nothing, so this is a free first lab.

**MEMORY HOOK: READ BEFORE YOU WRITE**

Before any `create`, `update` or `delete`, run the matching `list` or `show`. Before any PowerShell `New-`, `Set-` or `Remove-`, run the matching `Get-`. You learn the field names, and you catch the wrong-subscription mistake before it costs anything.

BEFORE YOU START

# How the lab repo works

One folder per ticket. Four commands. Cleanup you can verify.

The labs come with this book as **az104-field-labs.zip**, a second file in your download. They are also on GitHub at **github.com/azureawscert-a11y/az104-field-labs**. Every lab folder has the same core files (a few add extras like `policy.json` or `revoke.sh`), so once you have done one, you know how to do them all.




The lab loop. deploy.sh stops on the first error and is safe to run again; verify.sh and cleanup.sh keep going and report every problem. After step 5, rebuild the lab by typing the commands yourself. That is where it sticks.

## Get the lab files

CLOUD SHELL (BASH) · RECOMMENDED

```
# 1. In Cloud Shell: Manage files > Upload > az104-field-labs.zip
unzip az104-field-labs.zip -d az104-field-labs && cd az104-field-labs
# or: git clone https://github.com/azureawscert-a11y/az104-field-labs.git && cd az104-field-labs
chmod +x labs/*/*.sh capstone/*.sh tools/*.sh
git config core.fileMode false   # clones only: keeps git pull working
EMAIL=you@example.com bash tools/setup-subscription.sh
bash tools/check-region.sh   # before VM labs
# no storage mounted? Cloud Shell forgets files: repeat the steps above each session
cd labs/01-onboard-user-rbac
cat README.md
./deploy.sh
./verify.sh
  PASS  user is a member of grp-finance-readers (true)
  ...
  PASS  group has Reader on rg-az104-lab01 (1)
./cleanup.sh
```

**COST SAFETY, EVERY TIME**

- Set a **budget alert** first: US$20 a month, email at 50%, 80% and 100%. One command does it: `EMAIL=you@example.com bash tools/setup-subscription.sh` (or the portal steps in Chapter 2).
- Each lab uses one resource group, `rg-az104-labNN`. Cleanup deletes it.
- Run cleanup **the same day**. VMs, App Service plans and vaults bill by the hour.
- Azure itself may leave empty, free helper groups (`AzureBackupRG_<region>_1`, `NetworkWatcherRG`). They cost nothing; delete them with `az group delete -n <name> --yes` when you finish all labs.

**USE YOUR OWN SUBSCRIPTION**

A free account or Pay-As-You-Go subscription. Never your employer's production subscription. Labs 01, 02 and the capstone need Owner (they create role and policy assignments). Before the first VM lab, run `bash tools/check-region.sh`. New subscriptions are often blocked from small VM sizes or App Service S1 in busy regions; the script finds a region and size that work for you and every lab uses them.

BEFORE YOU START

# Inside every lab folder

The same core files every time. Learn the pattern once.

| File | What it does | What you learn from it |
|---|---|---|
| `README.md` | The ticket: who asked, why, acceptance criteria, interview answer | Read requirements like a professional |
| `deploy.sh` | Azure CLI in numbered steps, variables at the top, `set -euo pipefail` | The exact commands, in order |
| `verify.sh` | PASS/FAIL checks using `az ... --query`, curl and Network Watcher | How to **prove** a change, not hope |
| `cleanup.sh` | Removes locks, backup data and the resource group in the right order | Dependencies (locks, vaults) block deletes |
| `main.bicep` | Labs 05 and capstone: infrastructure as code | Read, change and deploy templates |

Every lab and the capstone were run end to end in a brand-new Pay-As-You-Go subscription in November 2026 (Azure CLI 2.91, Bicep 0.49), and the scripts were fixed wherever Azure had changed. Every script passes ShellCheck, and the included GitHub Actions workflow repeats that on each push. Azure keeps changing, so read each FAIL line carefully: it tells you what to look at.

MEMORY HOOK: THE TICKET MINDSET

**Who, what, where, prove, undo.** Who asked and who gets access. What exactly changes. Where (which scope, region, subnet). How you prove it worked. How you would undo it. Answer those five and you can handle any ticket, in the lab or at work.

ON THE EXAM

Test-takers report portal-style "lab" items (for example blob access and SSPR) and many questions that show Azure CLI or Bicep. Typing these commands yourself is the fastest way to recognise the right answer in a screenshot.

LAB MAP

# Twelve tickets, every exam domain

Do them in order the first time. Later, pick the ones that match your weak areas.

| # | Ticket | Outline areas practised | Time | Cost (deleted same day) |
|---|---|---|---|---|
| 01 | Onboard a new hire | Entra users and groups, built-in roles, assign at RG scope, interpret assignments | 20 min | Free |
| 02 | Enforce tags, stop deletes | Azure Policy, resource locks, tags | 30 min | Free |
| 03 | Share files with an auditor | SAS, stored access policies, keys, lifecycle management, secure defaults | 25 min | < $0.05 |
| 04 | Restore a deleted file | File shares, snapshots, soft delete for Azure Files | 25 min | < $0.05 |
| 05 | VM from Bicep with what-if | Interpret Bicep, modify it (add a data disk, add an NSG rule), what-if, deploy; create a VM | 30 min | $0.01–0.10/h |
| 06 | Resize and add a data disk | VM sizes, managed disks, run-command | 30 min | $0.02–0.15/h |
| 07 | Zero-downtime release | App Service plan, web app, HTTPS, deployment slots, slot settings | 30 min | ~$0.10/h |
| 08 | Registry to running container | ACR, ACI, Container Apps, sizing and scaling | 35 min | < $0.20 |
| 09 | Hub-spoke, NSG, prove it | VNets, peering, NSGs, effective rules, Network Watcher | 40 min | $0.02–0.20/h |
| 10 | Make storage private | Private endpoints, private DNS, storage firewall | 40 min | $0.03–0.12/h |
| 11 | Logs, KQL and alerts | Diagnostic settings, Log Analytics, KQL, alert rules, action groups, alert processing rules | 40 min | < $0.05 |
| 12 | Back up and restore a VM | Recovery Services vault, custom backup policy (14-day retention), backup now, restore disks | 90 min | ~$0.10/h |
| ★ | **Capstone:** clinic landing zone | All five domains in one handover-ready project | 2 to 3 h | ~$0.15/h |

Costs are rough USD estimates at the time of writing. The low end is `B2ats_v2`, the high end a D2-class v6 size if your subscription needs one. Prices change. Your budget alert is the real safety net.

LAB 01 · IDENTITY & GOVERNANCE

INC-1001 · labs/01-onboard-user-rbac

## Onboard a new hire

Priority 3 | 20 min | Free

> *"Priya starts Monday in Finance. She needs to see the finance resources in Azure but must not change anything." (Finance manager)*

### Acceptance criteria

- [ ] User created with a temporary password and forced change at first sign-in
- [ ] Security group `grp-finance-readers` with Priya as a member
- [ ] **Group** has **Reader** on `rg-az104-lab01` only
- [ ] Priya has no direct role assignment

### Try this after verify passes

Sign in as Priya in a private browser window. Open the resource group, then try to create a storage account in it. The portal refuses. That refusal is RBAC working.

KEY COMMANDS

```
DOMAIN=$(az rest --method get --url https://graph.microsoft.com/v1.0/domains \
    --query "value[?isDefault].id | [0]" -o tsv)   # your tenant's domain
az ad user create --display-name "Priya Sharma (lab)" \
    --user-principal-name priya.lab01@$DOMAIN \
    --password "$PASS" \
    --force-change-password-next-sign-in true
az ad group create --display-name grp-finance-readers \
    --mail-nickname grp-finance-readers
az ad group member add --group "$GROUP_ID" \
    --member-id "$USER_ID"
az role assignment create \
    --assignee-object-id "$GROUP_ID" \
    --assignee-principal-type Group \
    --role Reader --scope "$RG_ID"
```

**SAY IT IN THE INTERVIEW**

"I assigned the role to a group at the smallest scope that covers the need. When she changes teams I remove her from one group, and access reviews only look at groups."

END OF THE FREE SAMPLE

# You have read 21 of 259 pages.

This sample is the front matter, Chapter 2 and the first workplace lab of **AZ-104: The Azure Administrator Field Guide**. The lab files are free for everyone in the repo below.

**The full book adds:**

- All five exam domains, every command in Azure CLI and PowerShell
- 11 more workplace labs and a capstone project
- 307 original practice questions: 156 in the book, 151 in the Practice Pack that comes with it
- Land the Job: a sample resume, LinkedIn guide and interview answers
- Field Reference: portal paths, 28 real error messages with fixes, glossary and index

Lab files and the link to the full book:
github.com/azureawscert-a11y/az104-field-labs

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